Jiting Cai

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EDUCATION

Shanghai Jiao Tong University (SJTU), Shanghai, China *Aug. 2021 - Jun. 2025*
Bachelor of Engineering in Computer Science and Technology

- **Overall GPA: 4.12**/4.30 (**Ranking: 1**/101) — **Major GPA: 4.14**/4.30 (**Ranking: 1**/101)
- **Core Coursework and Grades**: Calculus (1) & (2) (A&A+), Linear Algebra (A+), Probability and Statistics (A+), Computer Organization (A+), Compiler Principles (A+), Computer Graphics (A+), Mathematical Foundations of Computer Science (A+), Experiments in Computer Organization (A+), Machine Learning (A+), Introduction to Data Science (A+)

Carnegie Mellon University, PA, U.S. *Aug. 2025 - Jun. 2027 (expected)*
Master of Science in Robotics

- **Overall GPA: 4.25**/4.00
- **Core Coursework and Grades**: 16-811: Math Fundamentals for Robotics (A), 16-831: Introduction to Robot Learning (A+), 16-720: Computer Vision (A+), 16-833: Robot Localization and Mapping (A+)

PUBLICATIONS & MANUSCRIPTS

1. **Take a Step Back: Rethinking the Two Stages in Visual Reasoning** [arXiv]
   Mingyu Zhang, **Jiting Cai***, Mingyu Liu, Yue Xu, Cewu Lu, Yong-Lu Li†. Proceedings of **ECCV 2024**.*
2. **Architect: Generating Vivid and Interactive 3D Scenes with Hierarchical 2D Inpainting** [arXiv]
   Yian Wang, Xiaowen Qiu*, Jiageng Liu*, Zhehuan Chen, **Jiting Cai**, Yufei Wang, Tsun-Hsuan Wang, Xian Zhou, Chuang Gan†. Proceedings of **NeurIPS 2024**.*
3. **LuciBot: Automated Robot Policy Learning from Generated Videos** [arXiv]
   Xiaowen Qiu, Yian Wang*, **Jiting Cai***, Zhehuan Chen, Chunru Lin, Tsun-Hsuan Wang, Chuang Gan†. **In submission**.*
4. **IPR-1: Interactive Physical Reasoner** [arXiv]
   Mingyu Zhang, Lifeng Zhuo*, Tianxi Tan, Guocan Xie, Xian Nie, Yan Li, Renjie Zhao, Zizhu He, Ziyu Wang, **Jiting Cai**, Yong-Lu Li†. Proceedings of **CVPR 2026**.*
5. **TartanMatch: Towards Universal Dense Correspondence Across Modalities** [arXiv]
   Hyeokjoon Kwon, **Jiting Cai***†, Ruogu Li*, Kritan Bhandari, Geethika Hemkumar, Parv Maheshwari, Yuheng Qiu, Yuchen Zhang, Sebastian Scherer, and Wenshan Wang. **In submission**.*
6. **Point It, Strike It: Direction-Conditioned Dynamic Manipulation of Deformable Linear Objects** [arXiv]
   Yi Yang, Xiang Fei*, Lehong Wang*, Zilin Dai**, Ruogu Li**, **Jiting Cai****, Liyao Chang, Xinyi Yang, Henry Kou, Ruijie Fu, Lu Li†, and Howie Choset. **In submission**.*

RESEARCH EXPERIENCES (SELECTED)

Project: Video Generation Model for Embodied Policy Learning Jan. 2025 - Now
MIT-IBM Watson AI Lab [website] & University of Massachusetts, Amherst [website]
Advisor: Prof. Chuang Gan [personal page]

· Proposed *LuciBot*, a pipeline that utilizes a general purpose video generation model to autonomously generate supervision signals for complex embodied tasks
· Leveraged video generation models to provide rich semantic guidance, including goal states, motion trajectories, contact information and grasping affordances, overcoming the limitations of code-based and VLM-based rewards

*These authors contributed equally to this work
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† Corresponding Author

- Generated complex scenes and task settings, including rigid tasks, deformable tasks, and articulated tasks, thereby validating the method's generalization across various task configurations
- Conducted rich experiments to test the method and contributed to the paper writing of the project as the co-first author [arXiv]

Project: Visual Reasoning Framework Design Apr. 2023 - Oct. 2024
MVIG Lab, RHOS Group, Shanghai Jiao Tong University [website]
Advisor: Prof. Yong-Lu Li [personal page]

- Divided the visual reasoning process into the symbolization and the reasoning stages and discovered that current visual reasoning techniques all lack generalization ability and would be restricted within a single task
- Analyzed differences between the symbolization stage and the reasoning stage of current visual reasoning model and recognized that the reasoning stage is more general compared to the symbolization stage, therefore introducing a new visual reasoning framework based on the analysis and applying the one-for-all reasoning network for the reasoning process
- Introduced the "approximation principle", which showcases that the ability of the reasoning module could be improved by adding more independent tasks that belong to different domains and modalities
- Contributed to the conclusion and paper writing of the project and published the paper at **ECCV 2024** as the co-first author [arXiv]

Project: Embodied AI and Scene Generation Mar. 2024 - Dec. 2024
MIT-IBM Watson AI Lab [website] & University of Massachusetts, Amherst [website]
Advisor: Prof. Chuang Gan [personal page]

- Proposed a zero-shot generative pipeline that creates diverse, complex, and realistic 3D interactive scenes to advance Embodied AI agents and Robotics research
- Proposed to leverage 2D prior from vision generative models to facilitate the 3D interactive scene generation process. Controlled this process by initializing from simulation-rendered image for hierarchical inpainting, ensuring consistent spatial features and controllable camera parameters and depth scale, facilitating accurate 2D to 3D lifting
- Improved the grounding of inpainting results for small objects by controlling their position, rotation, and scaling parameters and introduced the robots into the scene to perform diverse embodied tasks
- Wrote a paper on the findings and conducted extensive experiments during the rebuttal process, resulting in the paper's successful acceptance to **NeurIPS 2024** [arXiv]

Project: Multimodal Learning for Medical Diagnosis Jul. 2024 - May 2025
Department of Computer Science and Engineering, Shanghai Jiao Tong University [website]
Advisor: Prof. Yi Hong [personal page]

- Conducted a comprehensive survey of cutting-edge papers on the use of multimodal data to assist in the medical diagnosis of complicated diseases such as Alzheimer's disease
- Researched the currently mainstream medical image and text datasets, seeking compatible MRI, CT, and X-ray diagnostic images along with their corresponding textual diagnostic reports
- Leveraged existing Alzheimer's disease research reports to extract key information and applied large language models (LLMs) to generate a large-scale knowledge graph that can be embedded into the model, aiming to design a knowledge-driven multimodal diagnostic model
- Developed a new multimodal network that leverages a Mixture of Experts (MoEs) and a hierarchical feature extraction approach to progressively align and fuse data from image and text. Demonstrated through experiments that the network exhibits a decent performance in Alzheimer's diagnosis

Project: Applications of Differential Privacy Oct. 2022 - Mar. 2023
John Hopcroft Center for Computer Science, Shanghai Jiao Tong University [website]
Advisor: Prof. Liyao Xiang [personal page]

- Conducted a survey considering differential privacy as the crucial technique for privacy protection and network security
- Designed a special kind of differential privacy method that favors comparative efficiency
- Participated in mathematical analysis and conducting of the experiments

ACADEMIC SERVICE

- **Conference Reviewer: ICLR** 2025, **ICRA** 2027

HONORS (SELECTED)

Merit Student (top 5%) 2022, 2024
Shanghai Jiao Tong University
Academic Excellence Award Scholarship - A Level (top 3%) 2022
Shanghai Jiao Tong University
Yang Yuanqing Undergraduate Honors Scholarship (five per year) 2024
Department of Computer Science and Engineering, Shanghai Jiao Tong University
Outstanding Graduate 2025
Shanghai Jiao Tong University

SKILLS

Programming: C, C++, Python, PyTorch, MATLAB, Assembly Language, Verilog, LaTeX

Language: English (fluent), Mandarin Chinese (native), Cantonese Chinese (native)

- GRE **333+4.5**: Verbal 163 + Quantitative 170 + Analytical Writing 4.5
- TOEFL **108** : Reading 30 + Listening 28 + Speaking 23 + Writing 27

Teaching: Student Mentor of Shanghai Jiao Tong University

Music: Flute, Piano